Alberto E. Vilano Reyna

Senior QA Automation Engineer | AI Test Architecture

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PROFESSIONAL SUMMARY

Senior QA Automation Engineer and SDET wizard with 9+ years architecting production-grade test suites, developer tooling, and automated quality gates across high-reliability platforms. Proven track record leading squad quality strategy, pairing with product engineers on TDD and testability, and executing large-scale framework migrations (including migrating **200+ specs to Playwright**). Hands-on in backend/API automation (Java, TypeScript, Python), mobile verification (Appium), and AI-driven quality engineering (custom MCP servers connecting Atlassian Zephyr, automated spec generation). Experienced in production observability, log analysis, and incident triage using Elasticsearch and New Relic to safeguard transactional integrity and financial flows.

CORE TECHNICAL COMPETENCIES

AI Tools & Agent Orchestration:	**Claude Code, Cursor IDE**, Gemini, Agent Orchestration, **Agent Skill Authoring & Management** (SKILL.md), MCP Servers, Model Routing & Categorization, Agentic Learning & Process Auto-Improvement, LLM Evals, Synthetic Test Data
Automation & Mobile:	**Playwright**, Cypress, Appium, Selenium WebDriver, Cucumber (BDD), Grafana K6, Component & Visual Testing
Stack & Languages:	**TypeScript**, JavaScript (ES6+), Python, SQL, **Java (Spring Boot fundamentals)**, Angular basics, Bash/Shell
API & Performance Testing:	RESTful APIs, Microservices, **Postman, Bruno, JMeter**, Charles Proxy, Contract & Schema Validation
CI/CD, Cloud & Observability:	**GitHub Actions, Jenkins**, Docker, AWS (CloudWatch, S3, EC2), **New Relic / Datadog** (APM, Logs, Metrics), RCA, Log Matching, Triage Dashboards

PROFESSIONAL EXPERIENCE

Senior Software Development Engineer in Test — Coursedog, Inc.

Aug 2021 – Present | Remote

- **Framework Modernization & Stability:** Led the migration of **200+ specs from Cypress to Playwright**; implemented PR-level smoke gates across 1,200+ tests, reducing build flakiness by **>20%** and cutting CI execution cycle times by **30%**.
- **Team AI Enablement & QA Agents:** Authored and managed specialized QA AI Skills and agents for the engineering org to enforce code review conventions, automate PR quality gates, identify coverage gaps, and accelerate Playwright spec authoring and migration.
- **Incident Response & Observability:** Troubleshot production anomalies, conducted postmortem root-cause analyses (RCA), and analyzed distributed logs and metrics in Datadog and AWS CloudWatch to isolate backend regressions and verify transactional telemetry.
- **A11y Standards & Squad Enablement:** Spearheaded the org-wide introduction of accessibility (a11y) testing conventions and WCAG guidelines; paired with feature squads on TDD, predictable selectors best practice strategies, and established automated a11y checks in CI as standard release gates.
- **Visual Testing & Tooling Enablement:** Drove the end-to-end evaluation and rollout of visual testing frameworks (Applitools), leading tooling research, architecting POCs, authoring team conventions, and delivering documentation and training for org-wide adoption.

QA Engineer — Bold.org

Aug 2021 – Dec 2021 | Remote

- **Greenfield Quality Architecture:** Designed and deployed an end-to-end automation platform for a fast-growing financial/education application, establishing CI/CD quality gates using GitHub Actions.
- **Payment & Wallet Verification:** Automated comprehensive regression suites covering payment gateway integrations, user wallet balances, and PostgreSQL transactional state integrity.
- **Pipeline Optimization & Triage:** Configured concurrent test execution runners in CI/CD, boosting test run reliability and establishing sprint-level automation audits to accelerate defect triage.

Software Development Engineer in Test — Enroute / Integral Ad Science

Sep 2017 – Aug 2021 | Remote

- **Framework Modernization & Architecture:** Maintained and expanded custom test automation framework, migrating legacy codebases to Python 3, Java, and Selenium across distributed backend services and ad-measurement pipelines.
- **Partner & Integration Testing:** Architected automated verification suites covering core data and media integrations with partners including Twitch, Amazon, and Toyota, validating telemetry accuracy and tracking beacons.
- **Cloud & CI/CD Infrastructure:** Spearheaded the decommissioning of legacy on-prem test infrastructure by containerizing runners in Docker and transitioning CI/CD pipelines to AWS cloud environments using Jenkins.
- **Engineering Culture & Mentorship:** Held daily pair-programming sessions with dev pods to debug and write tests, championed shift-left unit/acceptance coverage, and mentored a cohort of engineering interns in QA automation.

EDUCATION & CREDENTIALS

The University of Texas Rio Grande Valley
Bachelor of Science in Engineering Physics

Certifications & Accreditations
AWS Certified Developer Associate • ISTQB Certified Tester (FL)